

5.1.3

1. $1^2 = 1(1+1)(2+1)/6$
2. This is a direct calculation.
3. Assume that, for some k , $1^2 + \cdots + k^2 = k(k+1)(2k+1)/6$.
4. That $1 + \cdots + (k+1)^2 = (k+1)(k+2)(2k+3)/6$.
5. We just substitute in $1 + \cdots + k^2 + (k+1)^2 = k(k+1)(2k+1)/6 + (k+1)^2$ by the inductive hypothesis and compute that this equals $(k+1)(k+2)(2k+3)/6$ as required.
6. Steps 1-5 contain everything required for a proof by induction.

5.1.4

1. $1^3 = (1(1+1)/2)^2$
2. Direct calculation
3. For some k , $1^3 + \cdots + k^3 = (k(k+1)/2)^2$.
4. That $1^3 + \cdots + (k+1)^3 = ((k+1)(k+2)/2)^2$.
5. As in the previous problem, we use the inductive hypothesis to show that $1 + \cdots + k^3 + (k+1)^3 = (k(k+1)/2)^2 + (k+1)^3$. The rest is algebra.
6. Steps 1-5 contain everything required for a proof by induction.

5.1.10

The first few values of this sequence are $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, and so on. We conjecture that the formula is $\frac{n}{n+1}$. Checking the first few values, as we've done, provides the base case. For the inductive step, we assume that $\frac{1}{1 \cdot 2} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$, and conclude that

$$\begin{aligned} \frac{1}{1 \cdot 2} + \cdots + \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} &= \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \\ &= \frac{n(n+2) + 1}{(n+1)(n+2)} \\ &= \frac{(n+1)^2}{(n+1)(n+2)} \\ &= \frac{n+1}{n+2} \end{aligned}$$

as required.

5.1.18

1. That $2! = 2^2$.
2. This is the calculation $2 < 4$.
3. That, for some k , $k! < k^k$.
4. That $(k+1)! < (k+1)^{k+1}$.
5. We note that $(k+1)! = (k+1)k!$. By the inductive hypothesis, this is less than $(k+1)k^k < (k+1)(k+1)^k = (k+1)^{k+1}$.
6. Steps 1-5 contain everything required for a proof by induction.

5.1.28

We prove this by induction. For the base case, we compute $3^2 - 7 \times 3 + 12 = 0$, which is nonnegative. For the inductive step, assume that $n^2 - 7n + 12$ is nonnegative for some n . Then $(n+1)^2 - 7(n+1) + 12 = (n^2 - 7n + 12) + (2n - 6)$. By the inductive hypothesis, the first term is nonnegative; since $n \geq 3$ by assumption, the second term is nonnegative.

5.1.32

By induction. Base case: take $n = 1$. Then $3|(1^3 + 2 \cdot 1)$. Inductive step: assume that $3|n^3 + 2n$. Now $(n+1)^3 + 2(n+1) = (n^3 + 2n) + (3n^2 + 3n + 3)$. By the inductive hypothesis, the first term is divisible by 3; the second term contains only numbers divisible by 3.

5.1.49

For there to be overlap between the two sets of horses in the inductive step, we must assume $k > 1$. What this proof actually shows is that, if you have a set of horses in different colors, you can find two horses in that set that are not the same color.

5.1.64

We know that, if p is a prime and $p|a_1a_2$, then $p|a_1$ or $p|a_2$. We will use this as the base case of a proof by induction. Assume that, for some n , we know that, for any $a_1, \dots, a_n$, if $p|a_1 \cdots a_n$, then $p|a_i$ for some i . Now choose $a_1, \dots, a_{n+1}$ and consider the case where $p|a_1 \cdots a_{n+1}$. Then either $p|a_1 \cdots a_n$, in which case we are done by the inductive hypothesis, or $p|a_{n+1}$, in which case we are also done.

5.1.54x

Base case: take $n = 1$. The only set of 2 integers not exceeding 2 is $\{1, 2\}$, and $1|2$. For the inductive step, assume that the claim is true for any set of $n + 1$ integers not exceeding $2n$ and take an arbitrary set of $n + 2$ integers not exceeding $2n + 2$. Now, either the set contains both $2n + 1$ and $2n + 2$ or it does not. If it does not, it contains at least $n + 1$ integers not exceeding $2n$, so we are done by the inductive hypothesis. In the case where B contains both $2n + 1$ and $2n + 2$, then either it contains $n + 1$ or it does not. If it contains $n + 1$, we're done, since $n + 1|2n + 2$. So let us assume it does not. Then consider $C = (B - \{2n + 2\}) \cup \{n + 1\}$. This contains $n + 1$ integers not exceeding $2n$, so it contains one that is divisible by another. If both of those are originally in B , we're done. The only alternative is for C to contain some element that divides $n + 1$ (since it contains no elements divisible by $n + 1$ except for $n + 1$ itself). This element must itself be in B , and, since anything that divides $n + 1$ divides $2n + 2$ as well, we are done.

5.1.62x

The base case can be checked directly, and the inductive step follows from noting that each additional line cuts all the others, so the $n + 1$ -th line increases the number of regions by $n + 1$.

5.1.76x

1. The base case can be checked by hand, but, for the inductive step to hold, we would want the inequality $\frac{1}{\sqrt{3n}} \frac{2n+1}{2n+2} < \frac{1}{\sqrt{3n+3}}$, which is in fact false for any n .
2. The inductive step here follows from the inequality $\frac{1}{\sqrt{3n+1}} \frac{2n+1}{2n+2} < \frac{1}{\sqrt{3n+4}}$. Since every term is positive, this inequality is true if and only if $\frac{1}{3n+1} \left(\frac{2n+1}{2n+2} \right)^2 < \frac{1}{3n+4}$. In fact, $\frac{1}{3n+1} \left(\frac{2n+1}{2n+2} \right)^2 - \frac{1}{3n+4} = -\frac{n}{4(1+n)^2(3n+1)(3n+4)} < 0$, so this second inequality is in fact true.

5.2.9

As in the hint, let $P(n)$ be the statement that $\sqrt{2} \neq n/b$ for any positive integer b . For the inductive step, assume that $P(k)$ is true for all $k < n$. We will try to prove $P(n)$. Assume for a contradiction that $\sqrt{2} = n/b$ for some b . Then $n = \sqrt{2}b$, so $n^2 = 2b^2$, so n is even, so $4n^2 = 2b^2$, so b is even, and $\sqrt{2} = (n/2)/(b/2)$. However, this contradicts $P(n/2)$. Therefore $P(n)$ is true and the proof by strong induction is complete. (Note that strong induction needs no base case.)

5.2.12

Let $P(n)$ be the claim that n can be written as a sum of distinct powers of 2. Assume that $P(k)$ is true for all $k < n$. We will prove $P(n)$.

If n is even, then, by $P(n/2)$, we can write $n/2 = 2^{a_1} + \dots + 2^{a_t}$ for some values $a_1, \dots, a_t$. Then $n = 2^{a_1+1} + \dots + 2^{a_t+1}$. If the a_i s are all distinct, so are the $a_i + 1$ s.

If n is odd, then, by $P(n-1)$, we can write $n-1 = 2^{a_1} + \dots + 2^{a_t}$, so $n = 2^0 + 2^{a_1} + \dots + 2^{a_t}$. Since $n-1$ is even, none of the a_i s are zero, so the powers are distinct.

5.3.4ab

1. $0, -1, -1, 0$

2. $1, 1, 1, 1$

5.3.8bd

1. $a_1 = 0, a_2 = 2, a_n = a_{n-2}$ for $n > 2$.

2. $a_1 = 1, a_n = (\sqrt{a_{n-1}} + 1)^2$ for $n > 1$.

5.3.12

We prove this by induction. The base case is $n = 1$ and we can check it by hand: $f_1^2 = 1$, while $f_1 f_2 = 1 \cdot 1 = 1$. For the inductive step, assume that $f_1^2 + \dots + f_n^2 = f_n f_{n+1}$ and compute

$$\begin{aligned} f_1^2 + \dots + f_n^2 + f_{n+1}^2 &= f_n f_{n+1} + f_{n+1}^2 \\ &= f_{n+1}(f_n + f_{n+1}) \\ &= f_{n+1} f_{n+2} \end{aligned}$$

as required.

Problem 1 of midterm 2 2016

We are given the base case; we only need the inductive step. Assume that $f(k)$ is irrational and assume for a contradiction that $f(k+1)$ is irrational. Then there exist integers a and b such that $a/b = f(k+1) = \sqrt{1 + f(k)}$. Or, squaring both sides, $a^2/b^2 = 1 + f(k)$. However, this implies that $f(k) = (a^2 - b^2)/b^2$, a rational number, which is a contradiction.

5.3.17x

The Euclidean algorithm will require n steps. In the base case, where $n = 1$, we simply run the Euclidean algorithm on $f_n = 1$ and $f_{n+1} = 1$. For the inductive step, assume that the Euclidean algorithm needs n steps to find the GCD of f_n and f_{n+1} . Then consider finding the GCD of f_{n+1} and f_{n+2} . Since $f_{n+2} = f_{n+1} + f_n < 2f_{n+1}$, the first step in the Euclidean algorithm is to go from (f_{n+1}, f_{n+2}) to $(f_{n+2} - f_{n+1}, f_{n+1}) = (f_n, f_{n+1})$. By the inductive hypothesis, n steps are required after this, for $n + 1$ total.